

Unified Systems Biology, Neuroanatomy & Memory Master Framework

COMPLETE SET OF 17 COMPUTATIONAL MODELS (IPO), LECTURE SLIDES & MEMORY SYSTEMS

Part 1: Molecular & Biological Computational Systems (IPO Models)

1. Ionic Bond Formation (e.g., Sodium Chloride)

Input 1:	Distance between Atom A and Atom B.
Input 2:	Number of electrons in the outer rings of Atom A.
Input 3:	Number of electrons in the outer rings of Atom B.
Input Device:	The outer electron rings of Atom B.
Processing:	Determining if atoms are close enough AND if the electron counts induce a reaction.
Processing Device:	The principle of energy efficiency (easier for Chlorine to gain one electron than lose seven).
Output:	Transfer of one electron leading to the formation of a Sodium Ion (Na^+).
Output Device:	Availability of Sodium ions for functions such as conducting electricity.

2. Ion Collisions

Input A:	Speed and direction of ion x BEFORE collision.
Input B:	Speed and direction of external object y BEFORE collision.
Input C:	Angle of collision.
Input Device:	Ion x.
Processing:	Laws of physics (motion and electromagnetic attraction/repulsion).
Processing Device:	Protons, neutrons, and electrons of ion x and object y.
Output:	Speed and direction of ion x AFTER collision.
Output Device:	Ion x.

3. Diffusion across a Membrane

Input A:	Concentration of an ion x on the INSIDE (Side A) of the cell.
Input B:	Concentration of an ion x on the OUTSIDE (Side B) of the cell.
Input Device:	The Barrier (Membrane) between Side A and Side B.
Processing:	An IF → THEN calculation: Is the concentration of ion x higher on Side A than Side B? If it can freely pass, net movement occurs.
Processing Device:	The internal energy in ions that causes random movement in all directions plus Entropy rules.
Output:	Net movement from the side with higher concentration to the side with lower concentration.
Output Device:	The Barrier (Membrane) resulting in effects such as electrical impulse conduction.

4. Membrane Lipid Molecule (Phospholipids)

Input A:	Distance of the hydrophilic head from water.
Input B:	Distance of the hydrophobic tail from water.
Input Device:	The head and the tail of the lipid molecule.
Processing:	An optimization calculation to find the angle of rotation that keeps the head as close as possible to water while keeping the tail as far as possible from it.
Processing Device:	The specific hydrophilic and hydrophobic properties of the head and tail.
Output:	The choice of angle of rotation and the physical act of rotating (or choosing not to).
Output Device:	The head and the tail of the molecule.

Note: The collective output creates the cell membrane, which allows oxygen to pass through but blocks K^+ ions.

5. Covalent Bond Formation (e.g., Water)

Input 1:	Distance between atoms.
Input 2:	Number of electrons in the outer rings of Atom A.
Input 3:	Number of electrons in the outer rings of Atom B.
Input Device:	The components of each involved atom (protons, electrons).
Processing:	Determining if atoms are close enough and if electron counts induce a reaction; energy efficiency.
Processing Device:	The components of each involved atom (protons, electrons).
Output:	Number of electrons shared and formation of a new molecule (H_2O).
Output Device:	The components of each involved atom (protons, electrons).

6. Physico-Chemical Properties of any Molecule

Input:	The specific combination of atoms of a specific molecule.
Input Device:	The components of each atom (protons, electrons, neutrons).
Processing:	Physical and Chemical laws governing interactions between components and neighboring molecules.
Processing Device:	The components of each atom (protons, electrons, neutrons).
Output:	Physico-chemical properties of the molecule as a whole or its variable components.
Output Device:	The components of each atom (protons, electrons, neutrons).

7. Binding of any 2 Molecules (e.g., 2 Amino Acids)

Input A:	Names and physico-chemical properties of the two molecules.
Input B:	Proximity/distance between the two molecules.
Input C:	Availability of energy close by for binding.
Input Device:	The components of each atom of these two molecules.
Processing:	Chemical laws governing whether binding is allowed, bond type, proximity, and energy requirements.
Processing Device:	The components of each atom of these two molecules.
Output:	Specific type of bond (Covalent, Ionic, or Hydrogen) and the list of new molecules formed.
Output Device:	The components of each atom of these two molecules.

8. Protein Folding

Input:	The specific sequence of amino acids (Primary Structure).
Input Device:	Components of each atom (protons, electrons) of the amino acids.
Processing:	Physical and Chemical laws causing bonds to form at specific unique points along the protein.
Processing Device:	Components of each atom (protons, electrons) of the amino acids.
Output:	Unique 3D shape (Secondary, Tertiary, or Quaternary structure).
Output Device:	Components of each atom (protons, electrons) of the amino acids.

9. "Lock and Key" Mechanism (Receptors)

Input:	The specific 3D shape of a molecule trying to attach.
Input Device:	The specific 3D shape of the receiving part of the receptor molecule.
Processing:	Physical and Chemical laws governing whether the shapes match properly.
Processing Device:	The specific 3D shape of the receiving part of the receptor molecule.
Output:	A signal sent by the receptor to another part of the cell (communication).
Output Device:	The receptor molecule itself.

10. Control of Hemoglobin (Hb) Levels (Negative Feedback)

Input 1:	Levels of hemoglobin or oxygen delivery to the Kidneys.
Processing:	The Kidneys determining if Hb is too low or too high.
Output 1:	Release of more or less Erythropoietin.
Input 2:	Erythropoietin sent through the blood to the bones.
Output 2:	Bone marrow manufacturing more or fewer Red Blood Cells to adjust levels.

11. Probability of Generating a Correct Protein Sequence

Input 1:	Number of variables at each sequence location (21 amino acid types).
Input 2:	The number of amino acids in a protein chain (N).
Processing:	Rules of probability (1 in N to the power of 21).
Output:	The resulting sequence probability calculations (e.g., 1 in 574^{21} for hemoglobin molecules).

12. Heat-Gated Ion Channel (Heat Receptors)

Input:	Temperature higher than 43°C or the presence of Capsaicin (a chemical).
Input Device:	Heat-gated ion channels designed with a "lock and key" mechanism for Capsaicin.
Processing:	Determining if the temperature is > 43°C or if the capsaicin concentration is high enough.
Processing Device:	Changes in the protein structure of the channel when heat or binding occurs, leading to the opening of the channel.
Output:	Positive ions flow into the cell.
Output Device:	The parts of the protein that open the channel, sending electricity through nerves so the person feels heat.

13. Mechanically-Gated Ion Channel (Pressure Receptors)

Input:	Mechanical force applied to the cell membrane.
Input Device:	The mechanically-gated ion channel itself, designed to respond to mechanical stretch.
Processing:	Detecting if force is applied, its duration, and how fast to adapt to continued force.
Processing Device:	The protein and membrane designed to be stretched sideways, which physically opens the channel at the center of the molecule.
Output:	Positive ions flow into the cell.
Output Device:	The parts of the protein that open the channel, which eventually send electricity through nerves to allow the person to feel pressure.

14. Na^+/K^+ Pump (Sodium-Potassium Pump)

Input A:	Concentration of Na^+ ions inside the cell.
Input B:	Concentration of K^+ ions outside the cell.
Input C:	Concentration of ATP inside the cell.
Input Devices:	Multiple binding sites on the protein molecule using a "lock and key" mechanism.
Processing:	Determining the amount of Na^+ and K^+ to transfer to maintain the Resting Membrane Potential (RMP) and concentration gradients.
Processing Device:	Structural changes in the pump's proteins triggered when Na^+ , K^+ , and ATP bind to them.
Output:	The speed of pumping Na^+ out of the cell and K^+ into the cell.
Output Device:	Parts of the protein molecule designed to move ions, making them available for functions like nerve impulses.

15. Voltage-Gated Na^+ Channel

Input:	Voltage inside the cell membrane becoming positive enough.
Input Device:	The voltage-gated Na^+ channel designed to detect voltage changes.
Processing:	IF voltage is positive enough, THEN open the gate; ions move via electrochemical gradient.
Processing Device:	The membrane and protein designed to open the ion channel when a specific voltage is detected.
Output:	Positive Na^+ ions flow into the cell.
Output Device:	The channel part of the protein; this inflow of ions alters the voltage across the neuron.

16. Voltage-Gated K^+ Channel

Input:	Voltage inside the cell membrane becoming positive enough.
Input Device:	Ion channel designed to detect voltage changes.
Processing:	IF voltage is positive enough, THEN open the gate; ions move via gradient tracking.
Processing Device:	Protein molecule designed to open the channel when a specific voltage is detected.
Output:	Positive K^+ ions flow to the outside of the cell.
Output Device:	The channel part of the protein molecule.

17. The Neuron as an IPO Computational Unit

Input Device: The **Dendrites** accept incoming sensory input signals.

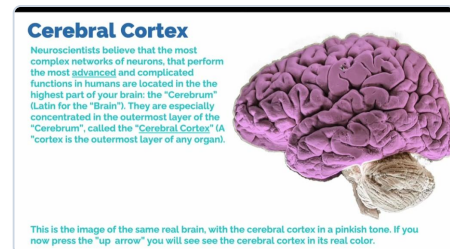
Processing Device: The **Soma (Body)** of the neuron performs integrated computational processing of incoming thresholds.

Output Device: The **Axon** carries out and propagates the outgoing action potential output signals.

Part 2: Neuroanatomy & Lecture Slides Framework

Cerebral Cortex Structure

- The complex networks of neurons that perform the most **advanced** and complicated functions are located in the "**Cerebrum**".
- They are concentrated in the outermost layer of the Cerebrum, called the "**Cerebral Cortex**".
- A cortex is structurally defined as the outermost layer of any organ.



2 Halves of the Cerebral Cortex

- The Cerebral Cortex is divided into **2 halves** (cerebral hemispheres).
- Each lobe is further subdivided into smaller areas, each carrying its own highly **specialized function**.

2 Halves of the Cerebral Cortex

This "Cerebral Cortex" is divided into 2 halves. Each has a different color here. Each lobe is further subdivided into smaller areas, each with its own specialized function.



Frontal Lobe Function

- The Frontal Lobe actively controls human executive **behavior and morality**.
- Clinical history shows that individuals injured in this specific area showcase drastically altered or **abnormal behavior**.

Frontal Lobe of the Cerebral Cortex

This controls behavior and morality.

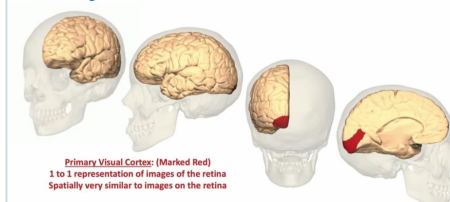
We know this because people injured in this area have historically shown abnormal behavior



Primary Visual Areas & Retina Mapping

- The **Primary Visual Cortex** (marked red) features a strict **1 to 1 representation of images of the retina** and remains spatially identical to layouts on the retina.
- A point of light drawing a line while moving from right to left on the retina stimulates a group of neurons that lie along a corresponding line in this primary cortex area.
- *Note: The main visual processing is here, NOT in the temporal lobe.*

Primary Visual Areas of the Brain



1. PRINCIPLES OF HUMAN MEMORY & LEARNING

- **Mechanism of Short-Term Memory:** Governed by temporary functional changes in neurons, specifically an increase in the **"firing rate"** (frequency of action potentials) running from certain neurons to adjacent ones. (e.g., repeating a 7-digit number triggers an immediate localized spike in firing activity).
- **Mechanism of Long-Term Memory:** Requires permanent, **physical structural changes**. This occurs by increasing the structural connection strength and the total number of physical connections (synapses) between neurons ("neurons that fire together, wire together").
- **The Transition from Short to Long-Term:** Structural rewiring takes time and sustained activation. Short-term memory acts as a **mandatory precursor**; information must first exist in short-term memory before it can be consolidated into long-term memory. However, consolidation is selective: short-term memory is only sometimes successfully converted.

2. NEURAL ENERGY CONSUMPTION & CELLULAR METABOLIC COST

- **The Cost of Neural Firing:** When neurons increase their firing rate to process or hold onto information, a flood of sodium (Na^+) and potassium (K^+) ions cross over the membrane via passive pathways.
- **The Role of the Pump:** To restore equilibrium and prepare the cells to fire again, the Na^+/K^+ pump must work overtime to drive those ions back against their concentration gradients.
- **ATP Consumption:** Because these pumps rely on active transport, a higher firing rate directly results in a surge of **ATP consumption**. Intense cognitive effort fundamentally raises the metabolic energy demands of your brain cells.

3. FOUNDATIONS OF NEUROANATOMY & CONSCIOUSNESS STUDIES

- **Synaptic Connections:** By definition, synapses are specialized communication junctions established between any two interacting neurons in the nervous system, enabling signal propagation throughout neural networks.
- **Exploring Consciousness via Edge Cases:** Analysing alternative perspectives is highly valuable in cognitive science to understand the true nature of consciousness:
 - *The consciousness of bats:* Exploring subjective experience ("What is it like to be a bat?" - Thomas Nagel).
 - *Philosophical zombies:* Hypothetical beings that behave exactly like humans but completely lack internal qualitative experiences.
 - *Patients in comas:* Examining clinical thresholds of awareness and neurological arousal parameters.